Renxiang Chu

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EDUCATION

Peking University **Beijing, China**

- Bachelor of Medicine in Basic Medical Sciences *Expected Jul. 2027*
  Dual Degree in Economics

Core Courses:

Medical Molecular and Cell Genetics, Organic Chemistry, Nervous System, Introduction to Computing (B), Principles of Economics, Probability and Statistics (B), Macroeconomics and Health

PUBLICATIONS

Chu, R.#, Sun, A.#, Qu, J., & Lu, M. (2026). ***Disentangling physiological heterogeneity in retinal aging using a deep learning-based biological age framework*** (https://doi.org/10.64898/2026.02.13.26346265). ***medRxiv*** (Preprint)

Chu, R.#, Sun, C.#, Zhang, Y.#, Sun, Z.#, Chen, Y., Shang, Y., Jiang, Y., & Lu, M. (2026). ***SciEngram: Evidence-grounded scientific memory for cumulative research reasoning***. Manuscript submitted to ***ICLR 2027***

Chu, R.#, Chi, X.#, Li, Z., Zhang, X., Xu, J., Wang, G., & Wei, X. (2026). ***HistoSeek links histology image recognition to teacher-authored formative guidance***. Manuscript under review at ***npj Digital Medicine***

Luo, T., **Chu, R.**, Shi, Z., Li, X., Li, Q., & Jin, W. (2026). ***One Patient, Two Futures: World Models for Clinical Trial Simulation***. Accepted at the ***WMHS Workshop at NeurIPS 2026***.

Equal contribution

RESEARCH EXPERIENCE

Peking University Basic Medical Sciences Innovative Talent Program **Position:** Research Trainee
Supervisor: *Tingting Li* *Sept. 2023 - Aug. 2024 | Beijing*

- Studied the computational characterization of biomolecular condensates in a bioinformatics research group.
- Developed R/Python pipelines to quantify condensate composition and phase-separation dynamics, integrating literature and experimental evidence to examine macromolecular assembly stability.

Peking University Institute for Advanced Clinical Medicine **Position:** Research Assistant
Supervisor: *Meng Lu* *Sept. 2024 - June 2027 | Beijing*

PROJECT: Retinal Aging and Biological Age Modeling

- Built a VisionFM-based retinal aging clock on 112,019 fundus images, achieving a mean absolute error of 2.49 years.
- Developed teacher-student residual learning to separate physiological aging signals from pathological variation; co-first-authored a medRxiv preprint (DOI: 10.64898/2026.02.13.26346265).

PROJECT: SciEngram - AI Co-Scientist Memory

- Designed SciEngram, a traceable cognitive-memory architecture and benchmark for long-term scientific reasoning across papers.
- Evaluated evidence attribution, cross-paper reasoning, conflict detection, and belief updating against long-context, RAG, and general-purpose memory baselines; co-first-authored a submission to ICLR 2027.

Peking University School of Basic Medical Sciences **Position:** Undergraduate Researcher (Part-time)
Supervisor: *Guangxi Wang* *Mar. 2026 - June 2027 | Beijing*

PROJECT: AI Co-Scientist for Cancer Biomarker Discovery

- Developed machine-learning and deep-learning multi-omics models and an automated workflow for cancer biomarker discovery.

PROJECT: HistoSeek - AI-Assisted Histology Education | Co-supervisor: *Xiaofan Wei*

- Designed HistoSeek, a teacher-governed platform for visual-localization practice and personalized feedback, and evaluated it in an undergraduate histology course.
- Co-first-authored the manuscript, currently under external review at npj Digital Medicine.

Northeastern University Khoury College of Computer Science **Position:** Intern (Remote)
Supervisor: *Wengong Jin* *Mar. 2026 - Dec. 2026 | Boston*

PROJECT: World Models for Clinical Trial Simulation

- Developed an action-conditioned world model using MIMIC and a private heart-failure dataset to simulate patient trajectories under candidate treatments for offline clinical-trial evaluation.
- Proposed RCT-based evaluation of world-model counterfactual reasoning and screened suitable trials; contributed causal tracing and model evaluation; co-authored an accepted WMHS Workshop paper at NeurIPS 2026.

INDUSTRY EXPERIENCE

Huagai Capital **Position:** Venture Capital Intern
Medical Devices & Technology Team *Sept. 2026 - Jan. 2027 | Beijing*

- Conducted investment-oriented industry research on brain-computer interfaces and AI for biology (AI4Bio), including AI-enabled molecular and drug discovery, virtual-cell models, laboratory automation, and DNA foundation models.
- Conducted on-site diligence of early-stage startups and interviewed founders to assess technical differentiation, product roadmaps, and commercialization potential.

LattixOrigin **Position:** Intern
Early-stage AI4S Startup *2026 - Present | Beijing*

- Contribute to research and model development for cross-scale human-brain foundation models, in parallel with work in the LM Lab.

LEADERSHIP & SERVICE

Team Leader, Peking University School of Basic Medical Sciences Social Practice Group

- Organized the Peking University School of Basic Medical Sciences social practice delegation to Suzhou under the theme "Exploring Development Achievements, Fostering the Spirit of Innovation," and served as team leader.
- Visited Soochow University and Kunshan Biomedical Industrial Park and met with Synermore Biologic Co., Ltd.

Bass-Baritone Vocalist, Peking University Health Science Center A Cappella Ensemble

- Performed bass and baritone parts in multiple ensemble performances.

SKILLS

- **Skills:** *MS Office (Proficient); Python/R (Working Knowledge);AI Tools*
- **Tools & Platforms:** *Linux, AWS, Conda, Jupyter, NumPy, Pandas, LaTeX, Git, Markdown*
- **Languages:** *English (CET-6: 607; TOEFL: 106); Mandarin Chinese (Native)*